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Education & Research Experience

- 2023–Present **PhD in Computer Science**
ENS, PSL University & INRIA, Paris. Supervisors: Dr. Ana Bušić, Prof. Jean-Michel Fourneau
Thesis: Performance Paradoxes and Admission Control in Stochastic Matching Models
- 2020–2023 **MSc in Mathematics**
Shandong University of Technology. Supervisor: Prof. Jianfeng Wang
Thesis: On the Matrix-Tree Theorems of Graphs. Defended: May 2023.
- 2015–2019 **BSc in Mathematics and Applied Mathematics**
Shandong University of Technology

Publications

- 2026 **Matrix-Tree Theorems for complex unit gain networks**
S. Li, J. Wang, Y. Li, M. Dehmer, M. Brunetti. Journal of Computational and Applied Mathematics, 488(2026) 117807.
- 2026 **A graph discretization of vector Laplacian**
S. Li, L. Lu and J. Wang. Discrete Applied Mathematics, 379(2026), 446-460.
- 2026 **Performance paradox in matching systems**
S. Li, A. Busic and J.M. Fourneau. ROADEF2026 Tours, France.
- 2025 **Performance paradox in matching models with abandonment**
A. Busic, J.M. Fourneau and S. Li. EAI VALUETOOLS, Glasgow, Scotland.
- 2025 **ORKM: An R package for online multi-view data clustering**
M. Yu, S. Li, G. Guo. Neurocomputing.
- 2025 **Performance paradoxes in matching systems are not that rare**
A. Busic, J.M. Fourneau, A. Lunven and S. Li. NETGCOOP, Bilbao, Spain
- 2023 **A graph discretization of vector Laplace operator**
S. Li, L. Lu and J. Wang. arxiv
- 2023 **Yet more elementary proof of Matrix-Tree Theorem for signed graphs**
S. Li and J. Wang. Algebra Colloquium, 30: 3(2023) 493-502.
- 2023 **On Graphs with Small Ranks: Old and New Results**
S. Li, Z. Stanic and J. Wang. Advances in Mathematics(China), 52(3) 385-405.

- 2023 **Matrix-Tree Theorem of digraphs via signless Laplacians**
S. Li, L. Lu, J. Wang and W. Wang. Linear Algebra and its Application, 658(2023) 32–48.
- 2017 **Equivalence Relation and Function**
S. Li and H. Li.
- Submitted **Extremal spectral radius of trees with given eccentric sequence**
S. Li, M. Yu and Y. Özer.
- Submitted **State space truncation and admission control for matching systems**
A. Busic, J.M. Fourneau and S. Li.

Presentations & Posters

- 2026 **Performance paradox in matching systems**
27ème Congrès Annuel de la Société Française de Recherche Opérationnelle et d'aide à la Décision (ROADEF2026) Tours, France
- 2025 **Performance paradox in matching models with abandonment**
18th EAI International Conference on Performance Evaluation Methodologies and Tools (EAI VALUETOOLS2025), Glasgow, Scotland
- 2025 **Performance paradoxes in matching systems are not that rare**
The 12th International Conference of Networks, Games, Control and Optimization (NETGCOOP2025), Bilbao, Spain
- 2025 **Performance paradoxes in matching systems are not that rare,**
LINCS Annual Workshop with the Scientific Committee, Palaiseau, France
- 2024 **A Brief Survey on the Matrix-Tree Theorems of Graphs,**
Argo Seminar Paris, France
- 2023 **On the Matrix-Tree Theorems of Graphs**
The Third SDUT Postgraduate Academic Forum, Zibo, China
- 2022 **On graphs with small ranks: old and new results,**
The 10th Int'l Combinatorics and Graph Theory Conference (CMS-CGT2022), Harbin, China
- 2022 **A Survey on Matrix-Tree Theorem of Graphs**
The 9th Int'l Symposium on Graph Theory and Combinatorial Algorithms (ORSC-GTCA2022), Hangzhou, China

Academic Service

Editor/Reviewers

- 2026-Present Mathematical Reviews(MathSciNet)
- 2025-Present Zentralblatt MATH(zbMath)

2025-Present Journal of Advances in Engineering and Technology

Peer review

Jrnl. Discrete applied mathematics / IEEE internet of Things Magazine / Asian Journal of Mathematics and Computer Research / Mathematics and Computer Science / Journal of advances in mathematics and computer science

2025 Confs. ICECET / GPS / ICIC / ACDSA / ICECCME

Professional Memberships

2023-Present The International Linear Algebra Society(ILAS)/ Société Mathématique de France(SMF)/ Société informatique de France(SiF)/Société de Mathématiques Appliquées et Industrielles (SMAI)/ Laboratory for Information, Networking and Communication Sciences(LINCS)

Brief Profile

I am a Ph.D. candidate in the ARGO team at Inria Paris and the Département d'Informatique de l'ENS (DI ENS, Université PSL), advised by Dr. Ana Bušić and Prof. Jean-Michel Fourneau. My research focuses on stochastic matching models, at the intersection of algorithms, optimization, and learning theory. Before my doctoral studies, I earned an M.Sc. in Graph Theory, where I worked on connections between graph structures and graph matrices, including the Matrix-Tree Theorem, low-rank graph families, and the vector Laplacian on graphs, in collaboration with Prof. Jianfeng Wang, Dr. Ning Bian, and other researchers. I also hold a B.Sc. in Applied Mathematics, through which I developed a strong foundation in linear algebra, number theory, and modern algebra. Beyond my thesis, I am interested in connecting theoretical insights with practical problems in computer science and mathematics.